

The University of the South Pacific

School of Computing, Information & Mathematical Sciences

CS412 Artificial Intelligence

Lab 4 (worth 4%)

This assessment covers the following course learning outcomes partially:

CLO 1 – Evaluate machine learning algorithms and their limitations

CLO 3 – Apply machine learning algorithms to solve real life complex computing problems.

In this lab you will learn about Neural Networks and its implementation that you have learnt in your lecture week 5. Do the following questions:

- 1 - Develop backpropagation Neural Networks algorithm in Matlab (preferred). Do not use any builtin tool. Refer to lecture notes for pseudocode.
- 2 - Use $\eta = 0.1$.
- 3 - Determine optimal size of hidden layers empirically.
- 4 - Use the dataset from Lab 3 (logistic regression) to test your algorithm.
- 5 - One of the termination criteria should be success rate. Use 85%+ to terminate.
- 6 - Does NN algorithm give better solution than logistic regression?

BOK	Unsatisfactory (0%-49%)	Satisfactory (50% - 75%)	Good (76% - 100%)	Marks Allocated	% Marks Attained
Abstraction & Design	Unsatisfactory II. Irrelevant prototyping VIII. Poor architecture and design makes future updates hard to incorporate.	Satisfactory I. Analysis of requirements such as risk analysis & mitigation, prototyping, identification of true/false requirements etc. III. all relevant constraints and requirements are mentioned VI. Appropriate choice of the architecture and design	Good I. All satisfactory and ... II. Choice of architecture and design is based on robustness, scalability, evolution and long term management. III. Innovative or creative design of algorithms (efficiency, structure and data structures)	1	
Programming	Unsatisfactory I. Plagiarism II. Poor indentation, hard to read and follow the code III. Lots of bugs and/or errors IV. Program produces unexpected output VII. Hard coding of data in the program. VIII. Poor structure	Satisfactory I. Able to write a complex code for an open ended problem II. Use of highest standard of programming practices such as modularity with appropriate architecture, low	Good I. All satisfactory and... demonstrate very good programming skills.	1	

		coupling, high cohesion, standard docs (eg. Javadoc) etc. III. Appropriate testing is done.			
Data & Information Management	Unsatisfactory VII. Very limited or undesirable data has been collected IX. Poor or incorrect inference is derived from the collected/given data	Satisfactory VII. Appropriate data has been collected IX. logical inference is derived from the collected/given data	Good I. All satisfactory and III. Extensive analysis is done on the collected/given data. IV. Exceptional Inference is derived from the data.	2	
Sub Total & comments				4	